INTRODUCTION TO OCEAN MODELLING

Inside an ocean model

From source code to sub-grid physics: how ROMS is built, what it eats, and how it solves the equations

Rohith B · INCOIS · 27.10.2026

ROADMAP · 60 MINUTES

Six stops on the way through a model

1

Build & run

CPP switches, compiling, the run loop · 8 min

2

Inputs

Grid, initial state, surface and boundary forcing · 10 min

3

Grids

Arakawa C-grid, terrain-following s-levels · 10 min

4

Numerics

Finite volumes: a linear example, then the temperature equation · 15 min

5

Parameterizations

Mixing, mesoscale and submesoscale eddies, viscosity · 12 min

6

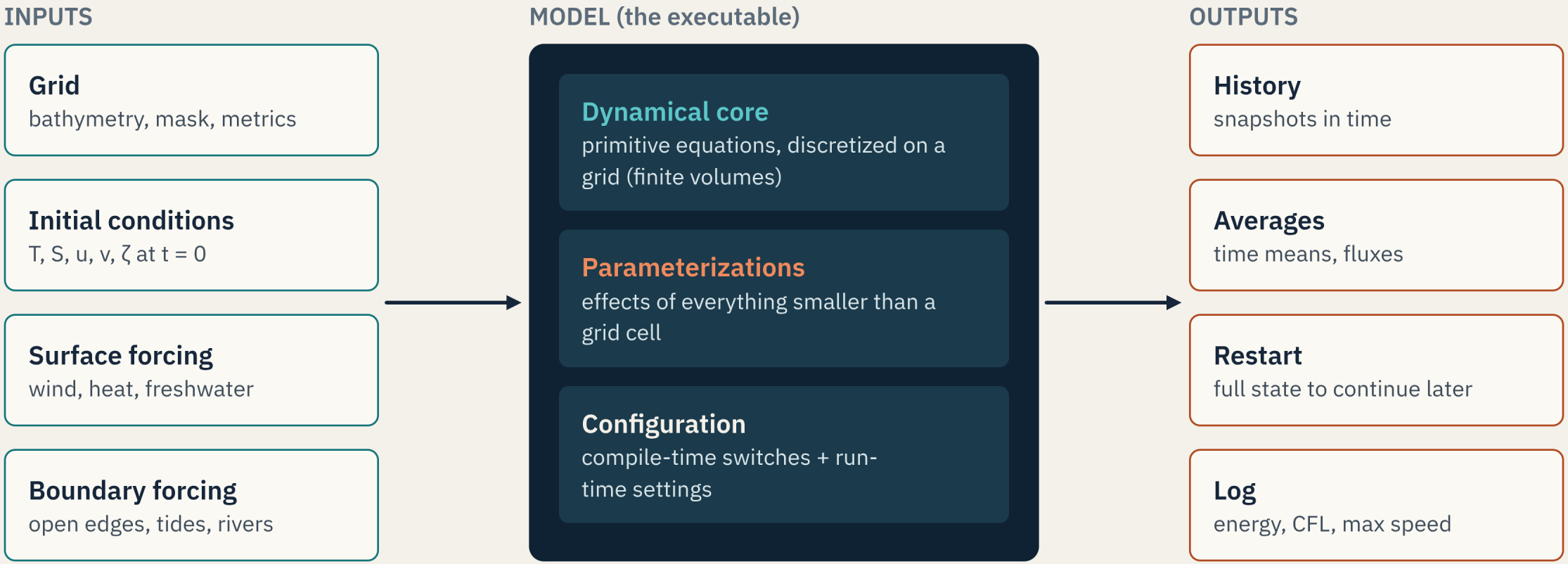
The core loop

One time step end to end, and sanity checks · 5 min

Hands-on: ROMS_class_2026.ipynb (build, run, analyse) · ROMS_input_preparation.ipynb (grid, IC, forcing)

THE BIG PICTURE

An ocean model turns inputs into a future state



All files are NetCDF; ROMS reads and writes them through the NetCDF-Fortran library.

THE EQUATIONS

ROMS solves the hydrostatic primitive equations

Momentum	$\partial u/\partial t + (u\cdot\nabla)u - fv = -(1/\rho_0)\,\partial p/\partial x + \partial/\partial z(K_m\,\partial u/\partial z) + D_u + F_u$
Hydrostatic	$\partial p/\partial z = -\rho\, g$
Continuity	$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$
Tracers (T, S)	$\partial T/\partial t + (u\cdot\nabla)T = \partial/\partial z(K_t\,\partial T/\partial z) + D_t + F_t$
State	$\rho = \rho(T, S, p)$

Orange terms = sub-grid effects, supplied by parameterizations (Part 5). The v-equation mirrors the u-equation.

Boussinesq
ρ varies by < 3 %, so ρ_0 replaces ρ except where multiplied by g

Hydrostatic
horizontal scales ≫ depth: vertical acceleration is dropped, w comes from continuity

Reynolds-averaged
equations hold for grid-cell means; eddy fluxes need closures

Shchepetkin & McWilliams (2005), Ocean Modelling 9; Haidvogel et al. (2008), J. Comput. Phys. 227

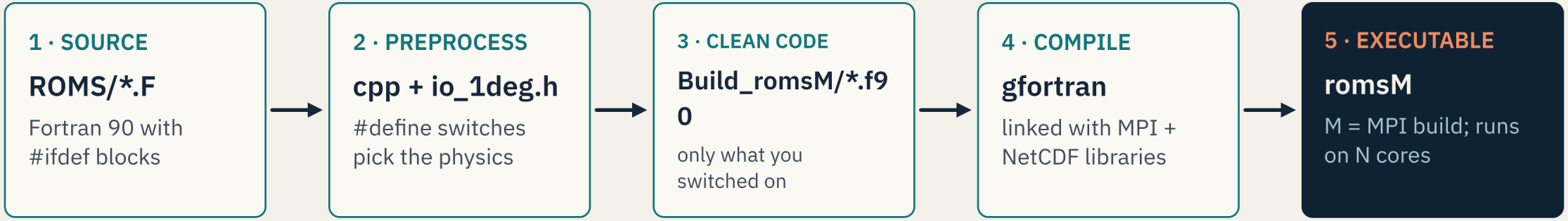
1

Build & run

From Fortran source to a running ocean: what you switch on, what you compile, what happens when you press enter

PART 1 · BUILD

Compiling: five stages from source to executable



In the Colab notebook

```
apt-get install libnetcdff-dev openmpi-bin
git clone github.com/myroms/roms
export ROMS_ROOT_DIR=/content
./build_roms.sh -j 2 > build.log 2>&1
```

WHAT build_roms.sh SETS

Variable	Meaning
ROMS_APPLICATION	IO_1DEG → reads io_1deg.h
MY_ROOT_DIR	where the ROMS source lives
USE_MPI, which_MPI	parallel build with OpenMPI
FORT	compiler: gfortran
USE_NETCDF4	finds NetCDF via nf-config

Build time on Colab's 2 cores: about 1.5 min. myroms.org/wiki/build_roms

PART 1 · CONFIGURE

Switches in the header decide which physics exists

#define	Adds to the model	Without it
UV_ADV, UV_COR	momentum advection, Coriolis	linear, non-rotating ocean
UV_VIS2 + MIX_S_UV	Laplacian viscosity	grid-scale noise, 4 m/s jets
TS_DIF2 + MIX_GEO_TS	tracer diffusion along z-levels	along-s diffusion mixes water masses on slopes
LMD_MIXING	KPP vertical mixing	no mixed layer, no convection
BULK_FLUXES + LONGWAVE_OUT	air–sea fluxes from ERA5	fluxes must be supplied ready-made
SOLAR_SOURCE	sunlight penetrates the water column	all heating in the top cell

Highlighted row: last year’s class header had the KPP block but never **#define LMD_MIXING**, so the run had no mixed layer at all.

Full list: ROMS/Include/cppdefs.h · myroms.org/wiki/cppdefs.h

PART 1 · RUN

A run reads settings, loops in time, writes files

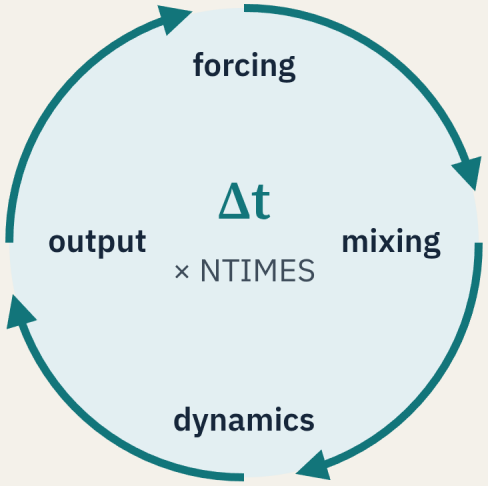
```
mpirun -np 2 ./romsM roms_io_1deg.in > roms.log
```

READS

roms_io_1deg.in
DT, NTIMES, viscosity, file names

varinfo.yaml
variable names, units, scales

grid · ini · frc .nc
the four inputs (Part 2)



WRITES

roms_his.nc
snapshots (NHIS)

roms_avg.nc
time means (NAVG)

roms_rst.nc + log
restart state, diagnostics

Class run: DT = 3000 s, 30 days = 864 steps, about 2 min on Colab

2

Inputs

A model is only as good as what you feed it: grid, initial state, surface forcing and open boundaries

PART 2 · INPUTS

Four inputs, four questions

Grid

Where is ocean, how deep, how big is each cell?

Sources: GEBCO, SRTM15+, Natural Earth coastline

`IO_1deg_grd.nc`

Initial state

What is the 3-D ocean doing at t = 0?

Sources: GLORYS12, HYCOM, World Ocean Atlas

`IO_IC.nc`

Surface forcing

What does the atmosphere do to the ocean?

Sources: ERA5, JRA-55, CFSR reanalyses

`ERA5.nc`

Boundaries

What does the rest of the ocean do at the open edges?

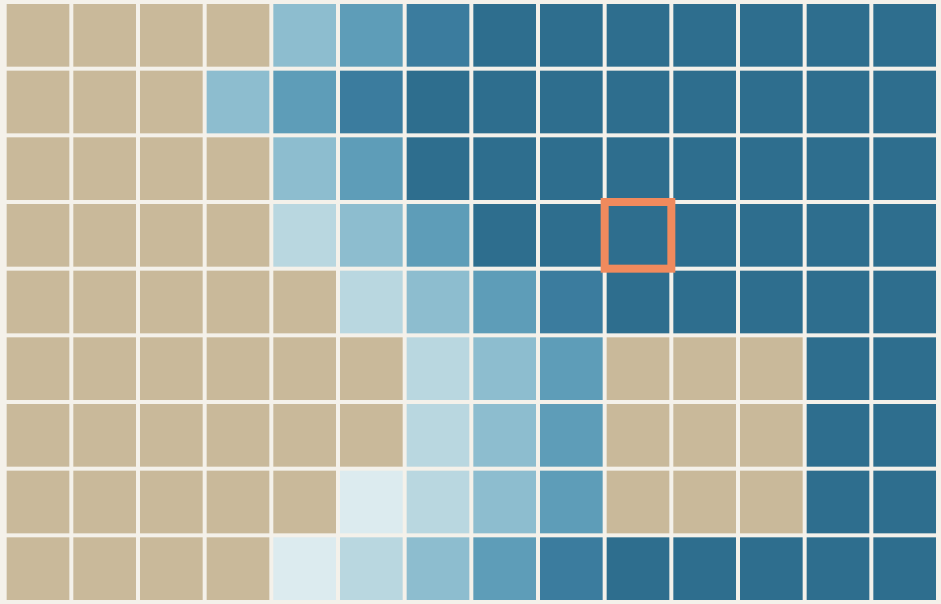
Sources: GLORYS or HYCOM, TPXO tides, river gauges

`none: closed walls`

Build them yourself with roms-tools: ROMS_input_preparation.ipynb

PART 2 · GRID FILE

The grid file: geometry the model never changes



land, mask = 0 shallow deep

Orange cell: one ρ-point. Its area is (1/pm)·(1/pn); its depth is h.

Variable	What it holds
h	water depth at the cell centre (m)
mask_rho, _u, _v, _psi	1 = ocean, 0 = land
pm, pn	$1/\Delta x$ and $1/\Delta y$ (m^{-1})
f	Coriolis parameter, $2\Omega \sin \varphi$
angle	rotation of grid axes from east
lon/lat _rho, _u, _v, _psi	positions of the four point types

Bathymetry is **smoothed** until neighbouring depths differ by less than ~40 %: $r_0 = |h_1 - h_2| / (h_1 + h_2) < 0.2$

Smoothing criterion: Beckmann & Haidvogel (1993), J. Phys. Oceanogr. 23

PART 2 · INITIAL CONDITIONS

From a reanalysis to the model's own levels

1 · SOURCE
GLORYS: T, S, u, v, ζ on 50 z-levels

→

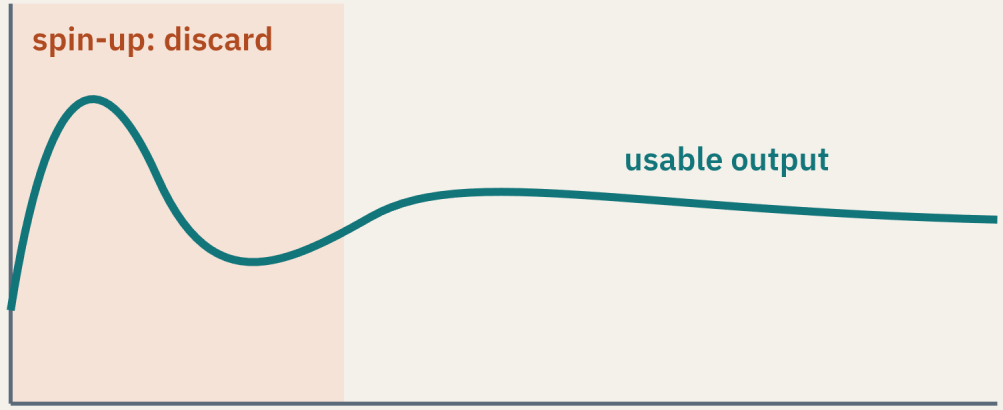
2 · FILL
land and below-seafloor gaps

→

3 · HORIZONTAL
interpolate to model ρ, u, v points

→

4 · VERTICAL
to s-levels; rotate u, v; compute ū, v̄



time → (domain-mean kinetic energy, schematic)

PITFALL WE HIT

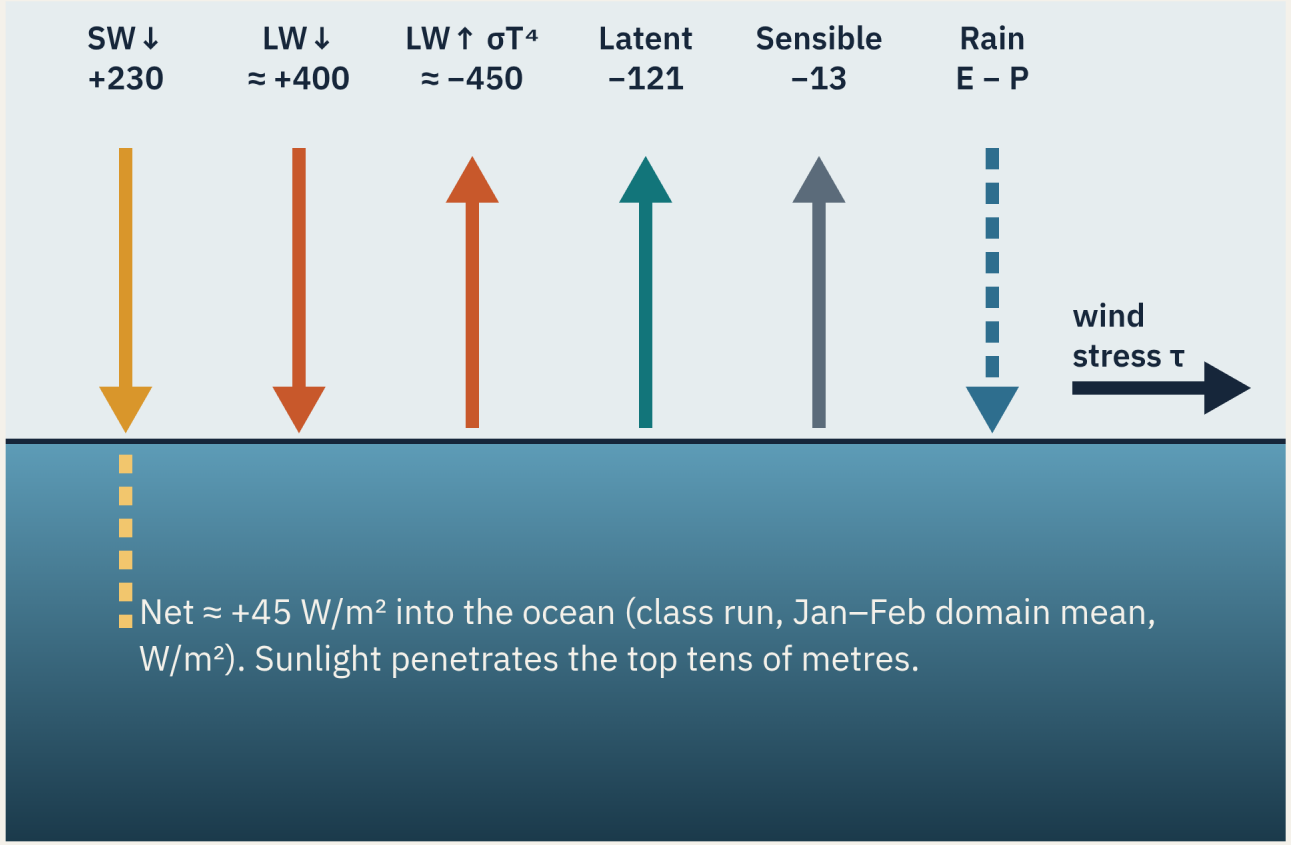
Saya de Malha bank: 35 m deep in GLORYS, 2,300 m in the smoothed 1° grid.

Copying the deepest value downward put 25 °C water at 1,000 m beside 10 °C water. ROMS blew up in 3 days. Fill gaps from neighbours **at the same depth**.

GLORYS12: Lellouche et al. (2021), Front. Earth Sci. · roms-tools: roms-tools.readthedocs.io

PART 2 · SURFACE FORCING

Bulk formulae turn weather into ocean fluxes



NET HEAT

$$Q = SW(1-\alpha) + LW\downarrow - \varepsilon\sigma T_s^4 - Q_e - Q_h$$

WIND STRESS

$$\tau = \rho_a \, C_D \, |U_{10}| \, U_{10}$$

SENSIBLE · LATENT

$$Q_h = \rho_a \, c_p \, C_H \, |U| \, (T_s - T_a)$$

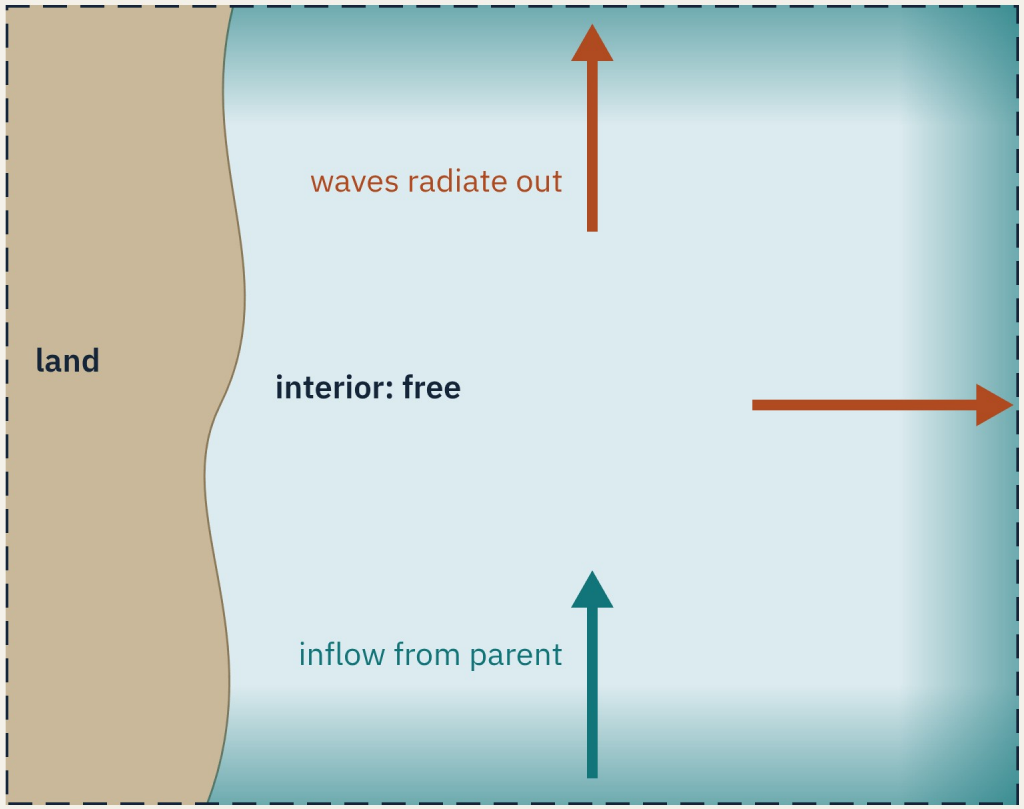
$$Q_e = \rho_a \, L_v \, C_E \, |U| \, (q_s - q_a)$$

The model's own SST enters every flux, so the ocean feels feedback: warmer water loses more heat.

COARE bulk algorithm: Fairall et al. (2003), J. Climate 16 · ERA5: Hersbach et al. (2020), QJRMS 146

PART 2 · BOUNDARY FORCING

Open boundaries: let waves out, let the ocean in



Shaded bands: nudging / sponge zones. Dashed edge: open boundary.

Fields	Typical condition
ζ, ū, v̄ (2-D)	Chapman for ζ, Flather for ū: tides and surges pass through
u, v, T, S (3-D)	radiation + nudging toward the parent model
near the edge	sponge: extra viscosity damps reflections
closed (our class)	wall: no flow through the edge

The problem is ill-posed: we prescribe values the interior also wants to set. Good conditions are a compromise between accepting inflow and transmitting outgoing waves.

Marchesiello, McWilliams & Shchepetkin (2001), Ocean Modelling 3 · Flather (1976) · Chapman (1985), J. Phys. Oceanogr. 15

3

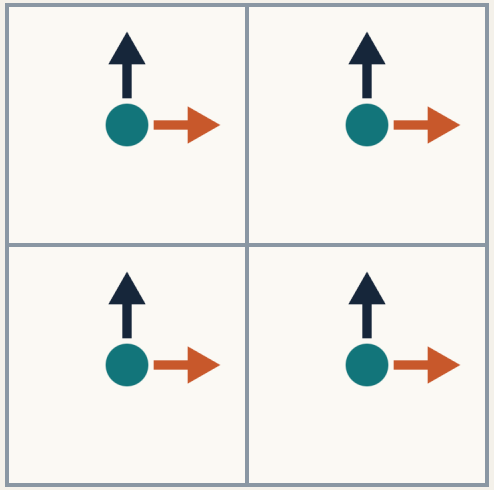
Grids

Where each variable lives: the Arakawa C-grid in the horizontal, terrain-following s-levels in the vertical

PART 3 · HORIZONTAL GRID

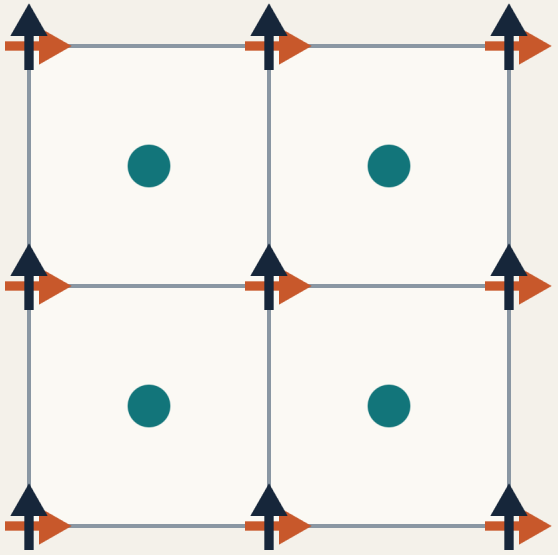
Arakawa grids: where u, v and T sit in a cell

T, S, ζ, p (ρ-point) — u (east–west velocity) | v (north–south velocity)



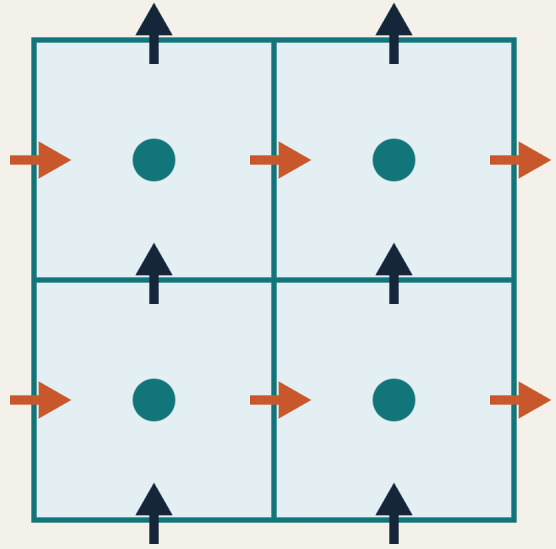
A-grid

all variables together; simple, but checkerboard noise in pressure



B-grid

u, v at corners; good Coriolis at coarse resolution (MOM)



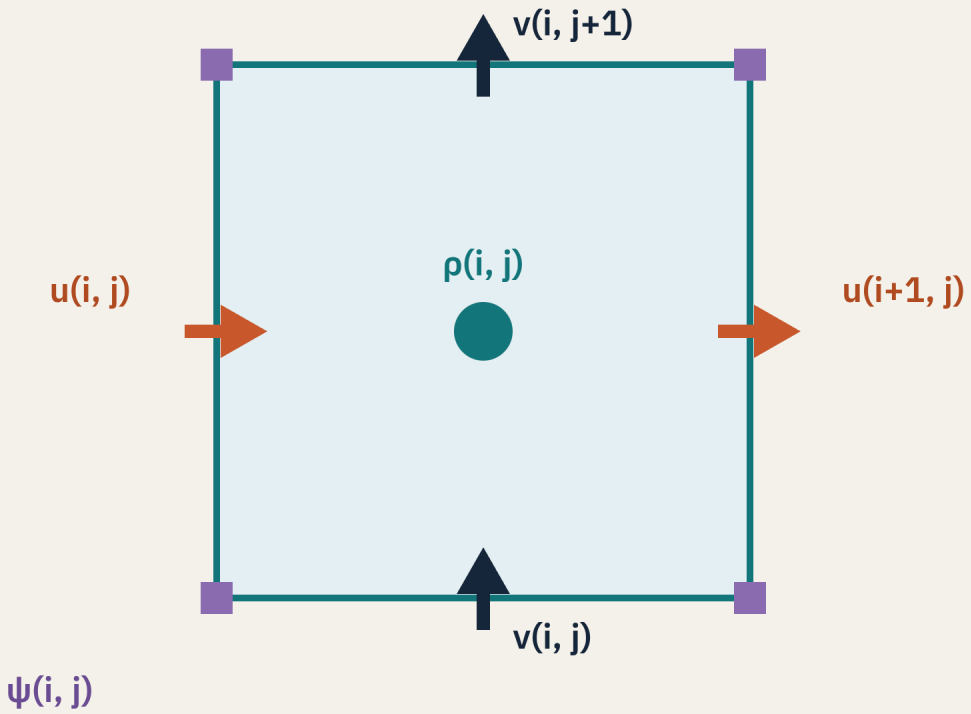
C-grid · ROMS

u, v on faces; best gravity waves when Δx is below the Rossby radius

Arakawa & Lamb (1977), Methods in Computational Physics 17 · also used by NEMO, MITgcm, CROCO, MOM6

PART 3 · THE ROMS C-GRID

One cell, four kinds of points



ρ-points: the cell volume
T, S, ζ, ρ are cell averages

u, v points: the faces
velocities carry the fluxes in and out, a finite volume by construction

ψ-points: the corners
vorticity and the no-slip / free-slip coast

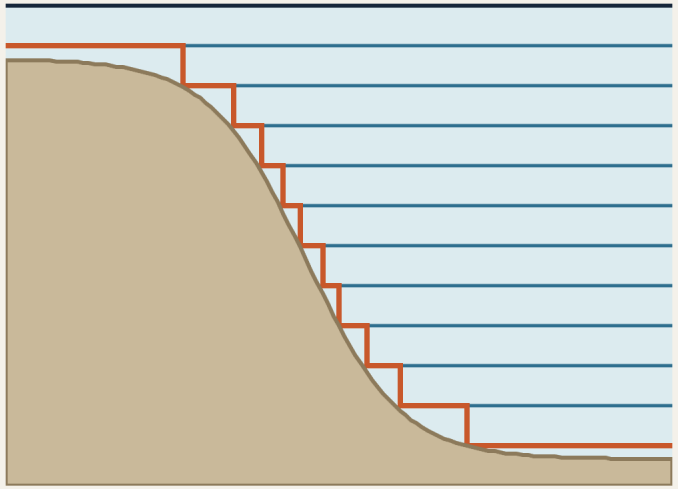
$$\partial \zeta / \partial t = -[U(i+1) - U(i)] - [V(j+1) - V(j)]$$

sea level changes only by what crosses the four faces (per unit area)

ROMS index convention: u(i, j) is the west face and v(i, j) the south face of ρ(i, j). myroms.org/wiki/Numerical_Solution_Technique

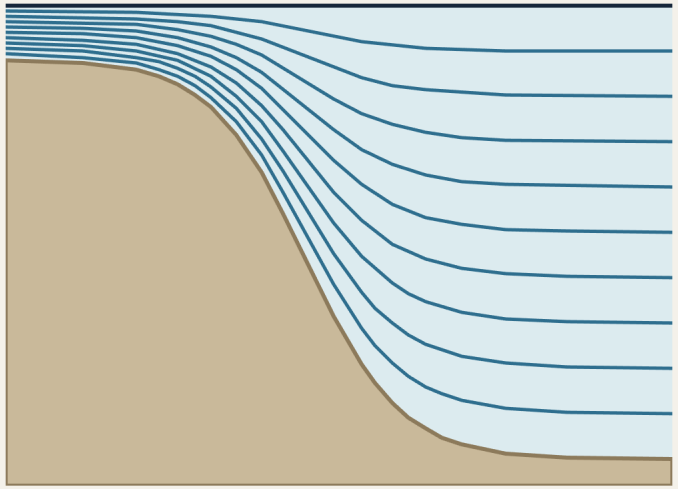
PART 3 · VERTICAL GRID

Three ways to slice the ocean vertically



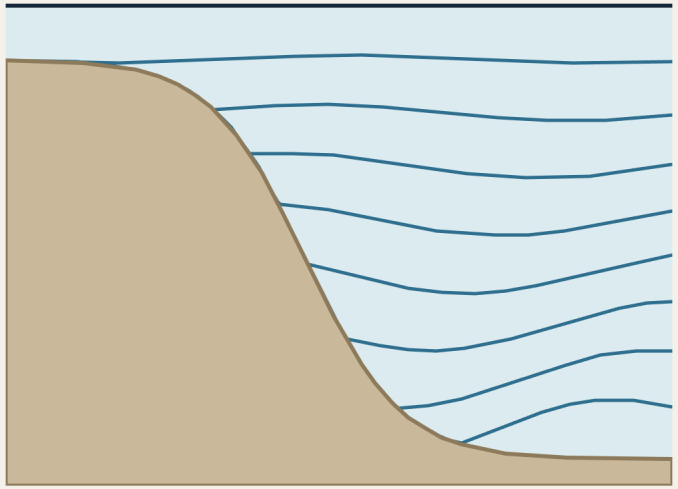
z-level

MOM, NEMO, MITgcm
+ clean pressure gradient
– staircase sea floor



Terrain-following (σ, s)

ROMS, CROCO, POM
+ smooth bottom, resolves shelves
– pressure-gradient error on slopes



Isopycnal (ρ)

HYCOM, MICOM (hybrid)
+ no spurious diapycnal mixing
– poor in the mixed layer

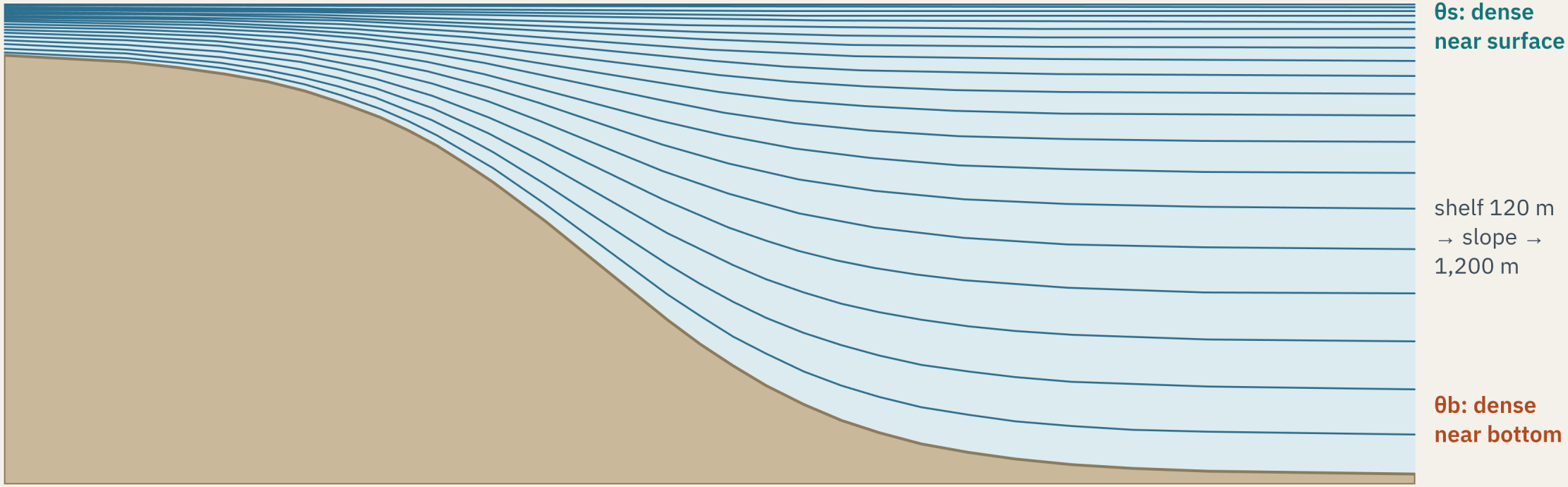
Griffies et al. (2000), Ocean Modelling 2: “Developments in ocean climate modelling”

PART 3 · S-COORDINATES

Stretching puts levels where the action is

$$z = \zeta + (\zeta + h) \cdot S,\ S = [hc\,\sigma + h\,C(\sigma)] / (hc + h)$$

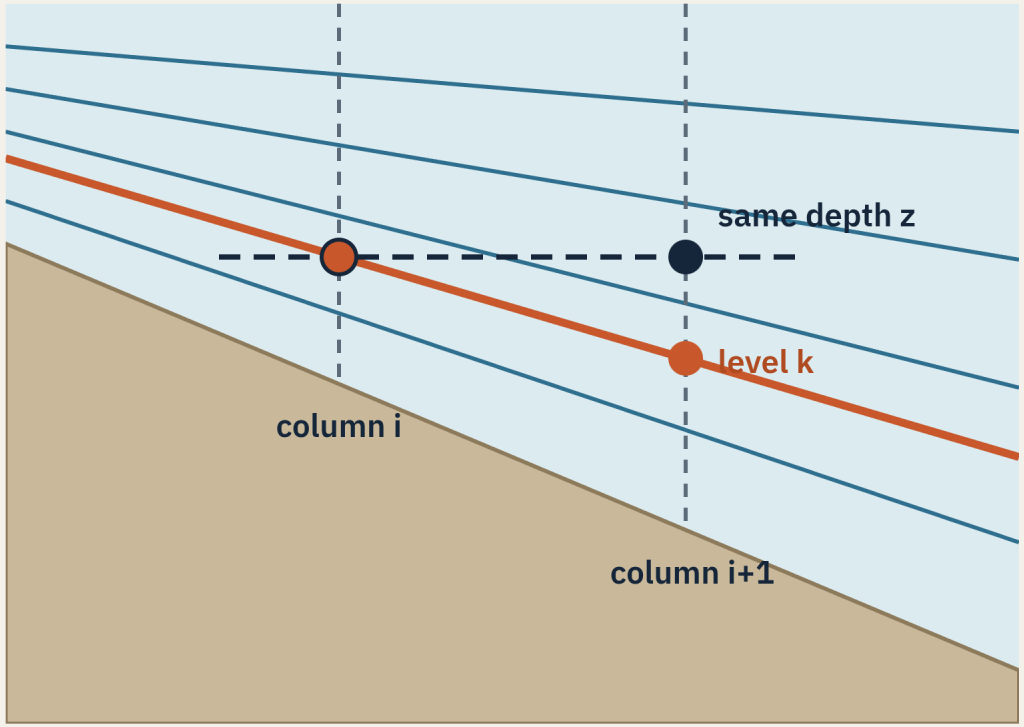
Class run: N = 40, θs = 5 (surface), θb = 2 (bottom), hc = 150 m · Vtransform 2, Vstretching 4



20 of 40 levels drawn · myroms.org/wiki/Vertical_S-coordinate · Shchepetkin & McWilliams (2005); Song & Haidvogel (1994), J. Comput. Phys. 115

PART 3 · THE PRICE OF FOLLOWING TERRAIN

Pressure-gradient error over steep slopes



Along level k the two points differ in depth: the model must subtract a large vertical pressure difference.

$$\partial p/\partial x|z = \partial p/\partial x|s - (\partial p/\partial z)(\partial z/\partial x)|s$$

two large terms, small difference: any error drives a spurious current

Metric	Measures	Keep below
rx0	bottom slope, $\lvert h_1-h_2\rvert$ / (h_1+h_2)	0.2
rx1	slope of s-levels vs their spacing	about 8

Our 1° grids	rx0	rx1
tutorial (croco_tools)	0.25	≈ 18
roms-tools	0.20	≈ 12

Remedies: smooth h · accurate pressure Jacobian (DJ_GRADPS) · fewer bottom-crowded levels

Haney (1991), JPO 21 · Beckmann & Haidvogel (1993), JPO 23 · Shchepetkin & McWilliams (2003), JGR 108

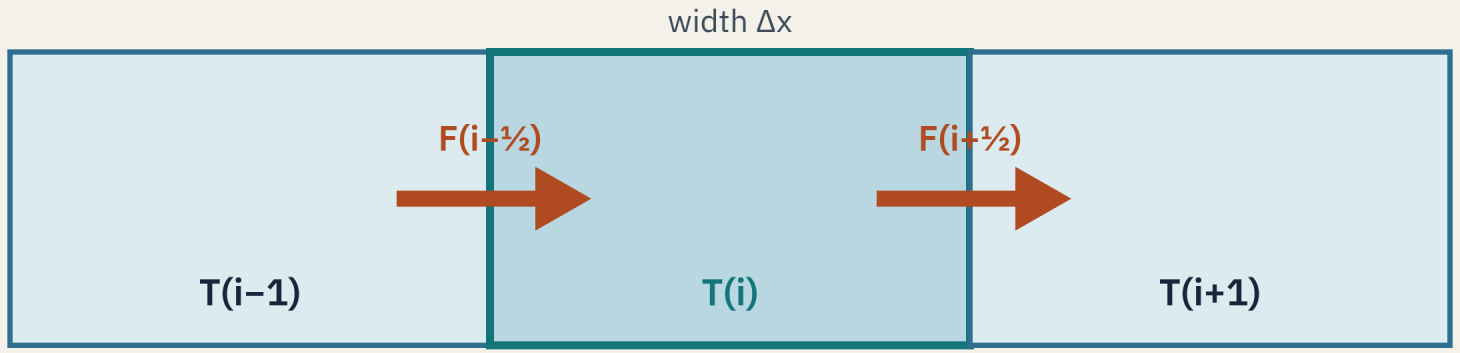
4

Numerics: finite volumes

How a differential equation becomes arithmetic: one linear example, then the full temperature equation

PART 4 · THE IDEA

Finite volume: count what crosses the walls



INTEGRAL FORM

$$d/dt \int T\, dV = -\oint F \cdot n\, dS$$

change inside = net flow through the walls

DISCRETE FORM

$$\Delta x \cdot d\bar{T}_i/dt = F(i-½) - F(i+½)$$

$\bar{T}_i$ is the cell average, not a point value

WHY OCEANOGRAPHERS LIKE IT

What leaves cell i enters cell i+1 exactly, so heat, salt and volume are conserved to machine precision.

LeVeque (2002), Finite Volume Methods for Hyperbolic Problems, Cambridge University Press

PART 4 · A LINEAR EXAMPLE

Advecting temperature in one dimension

1 Equation, constant current u > 0

$$\partial T/\partial t + \partial(uT)/\partial x = 0$$

2 Integrate over one cell and one time step

$$T_i^{n+1} = T_i^n - (\Delta t/\Delta x)\,[F(i{+}\tfrac{1}{2}) - F(i{-}\tfrac{1}{2})]$$

3 Choose the face value: upwind (take it from where the water comes)

$$F(i{+}\tfrac{1}{2}) = u\,T_i\,,\; F(i{-}\tfrac{1}{2}) = u\,T_{i-1}$$

= The update, with Courant number C = uΔt/Δx

$$T_i^{n+1} = T_i^n - C\,(T_i^n - T_{i-1}^n)$$

WORKED NUMBERS

u = 0.5 m/s, Δx = 10 km, Δt = 10,000 s

$$C = 0.5 \times 10^4 / 10^4 = 0.5$$

Upstream cell: 20 °C
This cell: 24 °C

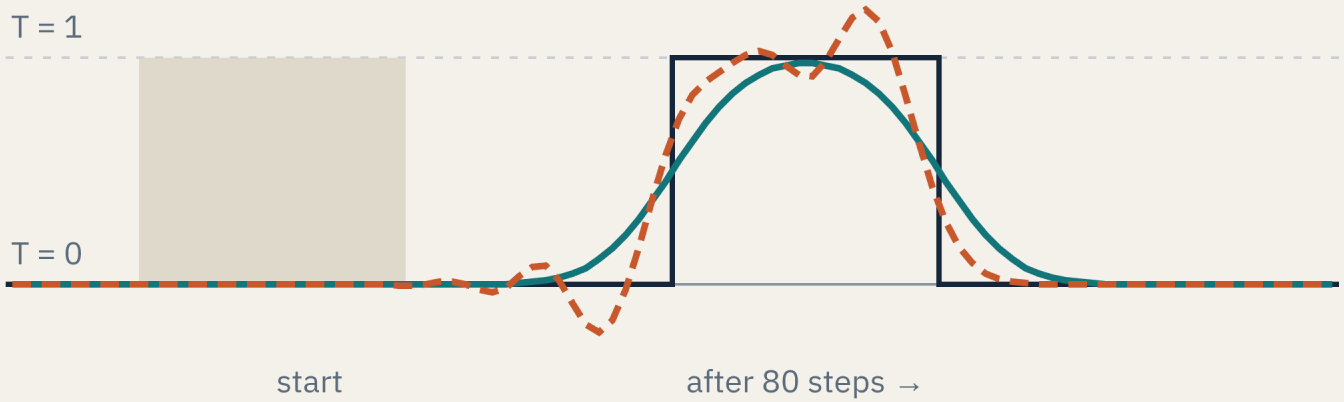
$$T^{n+1} = 24 - 0.5 \times (24 - 20) = 22\ ^\circ C$$

Half the cell's width of cooler water flowed in during one step.

First-order upwind (Courant, Isaacson & Rees 1952) · stable only if 0 ≤ C ≤ 1

PART 4 · WATCH IT RUN

Every scheme pays: smearing or wiggles



Exact: the pulse just moves

Upwind (1st order)

Lax–Wendroff (2nd order), dashed

Courant number 0.5, 100 cells, pulse moved 40 cells. Computed for this slide.

Numerical diffusion

Upwind never overshoots, but the sharp edges spread over ~20 cells: the scheme acts like extra mixing.

Dispersion

Higher order keeps edges sharp but creates new extremes: 1.21 and −0.21. Negative salinity is possible.

Godunov's theorem

No linear scheme above 1st order avoids new extremes. ROMS uses U3 horizontally, C4 vertically.

Godunov (1959) · Lax & Wendroff (1960), Comm. Pure Appl. Math. 13 · class roms_io_1deg.in: Hadvection U3, Vadvection C4

PART 4 · STABILITY

The fastest signal sets the time step

CFL CONDITION

$$C = c\,\Delta t / \Delta x \le 1$$

Signal (Δx = 111 km)	Speed c	Largest Δt
currents	~1 m/s	~30 hours
internal gravity waves	~3 m/s	~10 hours
surface gravity waves √(gH)	~200 m/s	~9 minutes

√(9.81 × 4000 m) ≈ 200 m/s. Running everything at 9 minutes would waste most of the computing on the barotropic mode.

MODE SPLITTING · ONE BAROCLINIC STEP

3-D baroclinic: 1 step, Δt = 3000 s



2-D barotropic: 60 steps of 50 s (NDTFAST = 60)

The 2-D equations have one level, so 60 cheap sub-steps cost far less than shrinking the whole 3-D step. Barotropic Courant number: 200 × 50 / 111,000 ≈ 0.09.

Courant, Friedrichs & Lewy (1928) · split-explicit scheme: Shchepetkin & McWilliams (2005), Ocean Modelling 9

PART 4 · FROM LINEAR TO REAL

The temperature tendency equation, term by term

$$\frac{\partial T}{\partial t} = -\nabla\cdot(uT) + \frac{\partial}{\partial z}\left(K_t \frac{\partial T}{\partial z}\right) + \nabla_h\cdot(A_h \nabla_h T) + \frac{1}{\rho_0 c_p}\frac{\partial I}{\partial z}$$

tendency	advection (resolved)	vertical mixing (KPP)	horizontal diffusion	penetrating sunlight

SURFACE BOUNDARY CONDITION

$$K_t \frac{\partial T}{\partial z}\Big|_0 = Qnet / (\rho_0 c_p)$$

The bulk fluxes from Part 2 enter here, through the top face of the top cell.

ORDER OF MAGNITUDE

100 W/m² into a 50 m mixed layer:

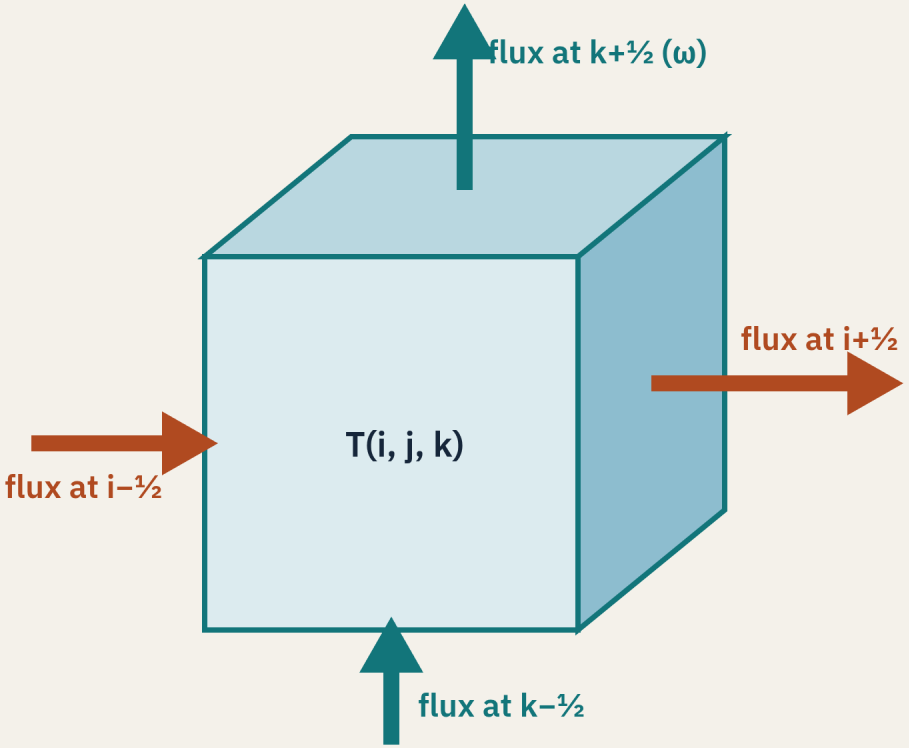
$$100 / (1025 \times 3985 \times 50) \approx 4.9 \times 10^{-7}\ K/s$$

≈ **1.3 °C per month**

Orange and gold terms come from parameterizations and forcing (Part 5). Salinity has the same form, with E − P at the surface.

PART 4 · SAME RECIPE, THREE DIMENSIONS

One ROMS tracer cell and its six faces



DISCRETE TRACER EQUATION

$$\partial(\Delta V\, T)/\partial t = -[UT](i{+}\tfrac{1}{2}) + [UT](i{-}\tfrac{1}{2}) - [VT](j{+}\tfrac{1}{2}) + [VT](j{-}\tfrac{1}{2}) - [WT](k{+}\tfrac{1}{2}) + [WT](k{-}\tfrac{1}{2}) + \mathit{mixing} + \mathit{sources}$$

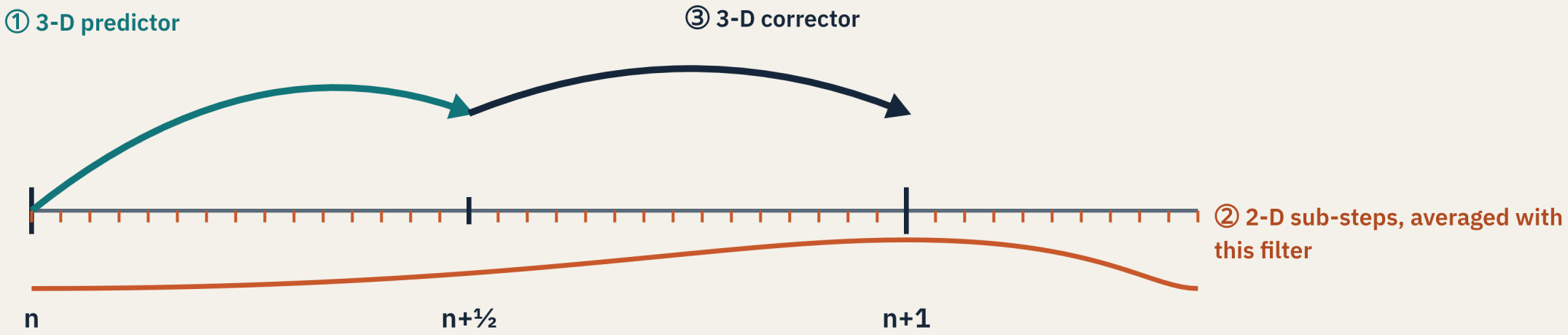
Symbol	Meaning in ROMS
ΔV = Hz/(mn)	cell volume; Hz = layer thickness, 1/m · 1/n = area
U = Hz u / n	volume flux through a u-face (m³/s)
V = Hz v / m	volume flux through a v-face (north–south faces)
W = ω / (mn)	flux through the sloping s-surface above and below

Face values of T come from the advection scheme. Vertical mixing is solved implicitly (a tridiagonal system per column), so large K_t never limits Δt.

Haidvogel et al. (2008), J. Comput. Phys. 227 · myroms.org/wiki/Numerical_Solution_Technique

PART 4 · TIME STEPPING

ROMS's split-explicit core: predict, sub-cycle, correct



3-D: LF-AM3
Leapfrog predictor + 3rd-order Adams–Moulton corrector for momentum and tracers

2-D: forward–backward
AB3–AM4 sub-steps for ζ and ū, v̄; a filter averages them to remove fast-mode aliasing

Coupling
Averaged 2-D transports replace the depth-mean of 3-D velocity: volume and tracers stay conserved

Shchepetkin & McWilliams (2005), Ocean Modelling 9, 347–404 · (2009), Handbook of Numerical Analysis 14

PART 4 · CHOOSING A SCHEME

Tracer advection options in ROMS

Scheme (.in keyword)	Order	Behaviour	Typical use
Centred (C2)	2	cheap; wiggles near fronts	tests
Centred (C4)	4	accurate, little diffusion, some wiggles	vertical (our run)
Upstream-biased (U3)	3	built-in scale-selective damping	horizontal (our run)
Split U3 (SU3)	3	damping along isopycnals, not across	steep topography
MPDATA, HSIMT	2–3	positive-definite, no negative values	biology, sediment

Every choice trades sharpness against spurious mixing; the hidden diffusion of a scheme can exceed the explicit diffusivity you set.

Shchepetkin & McWilliams (1998), Mon. Wea. Rev. 126 · Marchesiello, Debreu & Couvelard (2009), Ocean Modelling 26

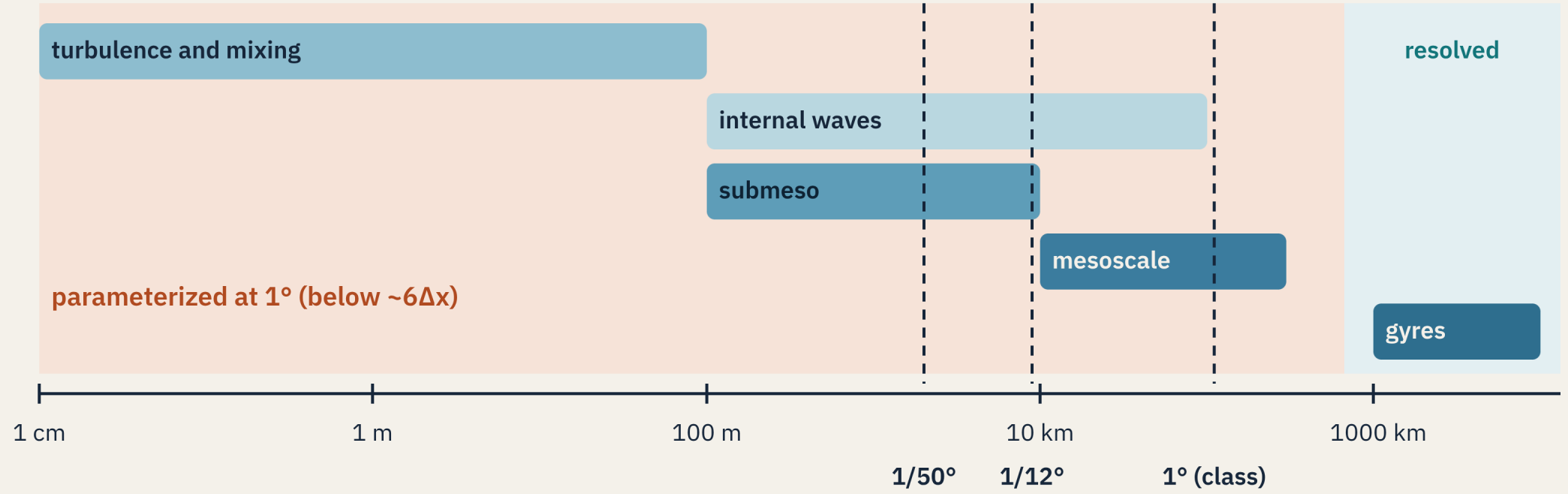
5

Parameterizations

What the grid cannot see: turbulent mixing, mesoscale and submesoscale eddies, and the viscosity that keeps it all stable

PART 5 · SCALES

Nine decades of motion, one grid spacing

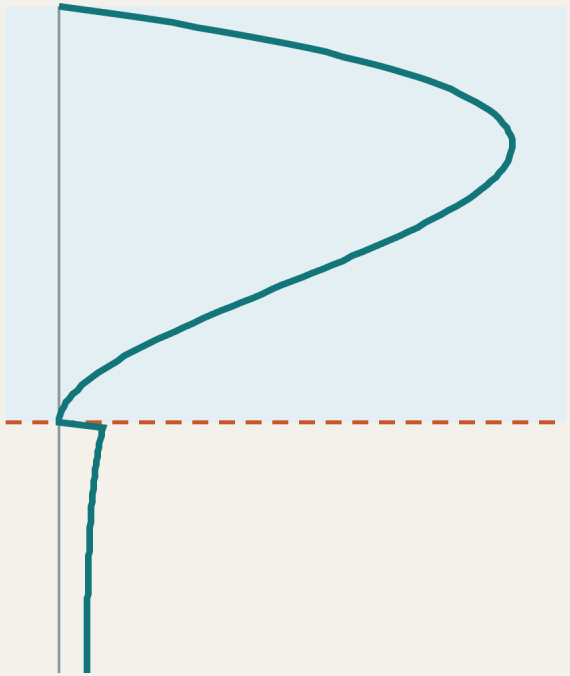


Mesoscale eddies are set by the first baroclinic Rossby radius: about 200 km near the equator, under 20 km at high latitudes.

Chelton et al. (1998), J. Phys. Oceanogr. 28 · effective resolution ≈ 6–10Δx: Soufflet et al. (2016), Ocean Modelling 98

PART 5 · VERTICAL MIXING

KPP: a shaped diffusivity in the boundary layer



INSIDE THE BOUNDARY LAYER

$$K(\sigma) = h \cdot w(\sigma) \cdot G(\sigma),\ G \approx \sigma(1-\sigma)^2$$

σ = depth / h; w = turbulent velocity scale from wind stress and buoyancy flux

Depth h
where the bulk Richardson number reaches a critical value (≈ 0.3)

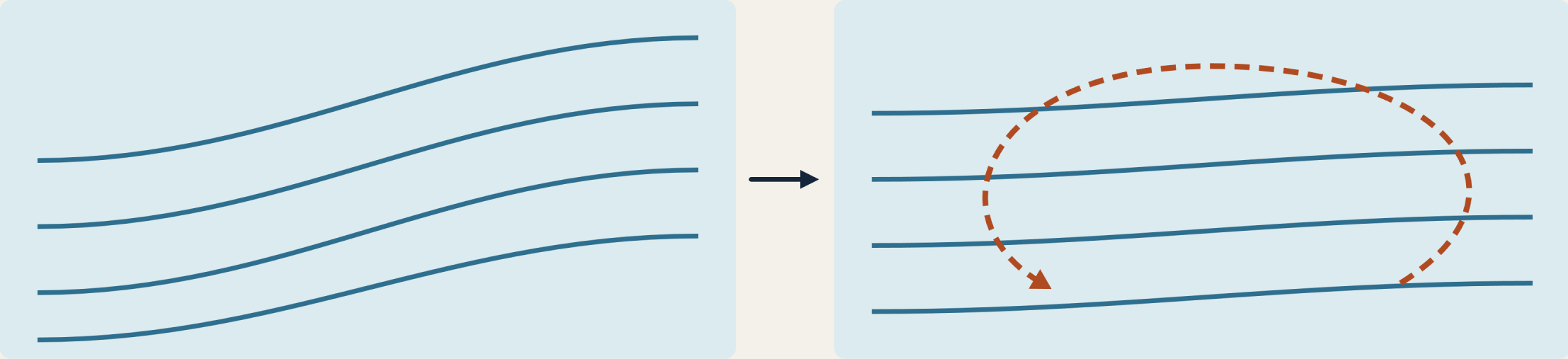
Non-local term
convection carries heat even against the local gradient

In ROMS: #define LMD_MIXING (+ LMD_SKPP, LMD_BKPP). Output Hsbl: 50–100 m in the winter Arabian Sea in our run. Alternatives: GLS (k–ε, k–ω), Mellor–Yamada 2.5.

Large, McWilliams & Doney (1994), Rev. Geophys. 32 · Umlauf & Burchard (2003), J. Mar. Res. 61

PART 5 · MESOSCALE EDDIES

Eddies flatten density surfaces: Gent–McWilliams



Unresolved front: steep isopycnals, stored potential energy

Eddy-induced overturning flattens them, releasing energy

$$u^* = -\partial(\kappa S)/\partial z \,,\; w^* = \nabla_h \cdot (\kappa S)$$

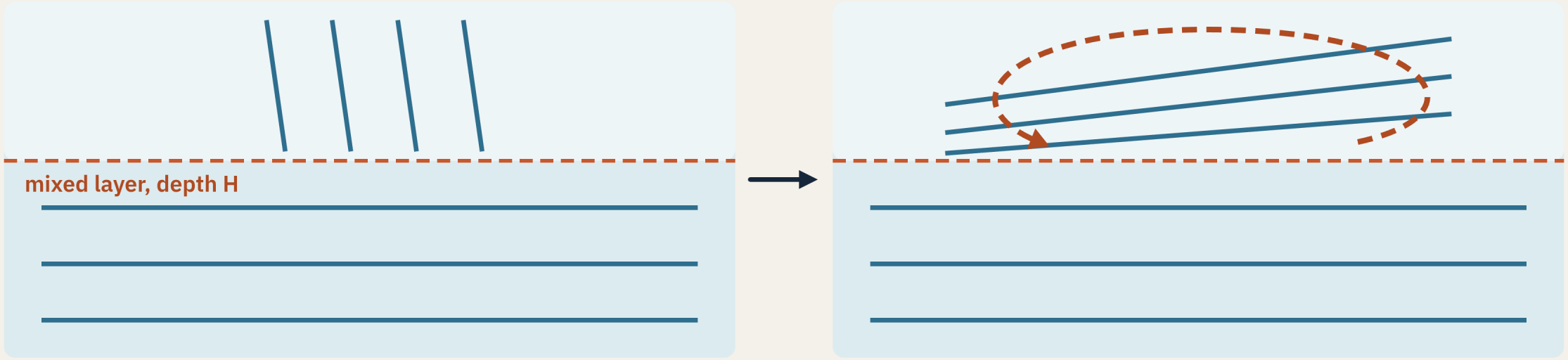
S = isopycnal slope, κ ≈ 300–1000 m²/s; plus Redi diffusion along isopycnals

Needed when Δx is larger than the Rossby radius: standard in climate models (NEMO, MOM, POP). Regional ROMS runs are usually eddy-permitting instead.

Gent & McWilliams (1990), J. Phys. Oceanogr. 20 · Redi (1982), J. Phys. Oceanogr. 12

PART 5 · SUBMESOSCALE

Mixed-layer eddies restratify the surface ocean



Front in the mixed layer: 0.1–10 km wide

Slumping shoals the mixed layer within days

$$\Psi = C_e \, (H^2 / |f|) \, \mu(z) \, \nabla b \times \hat{z}$$

overturning streamfunction from the horizontal buoyancy gradient ∇b; $C_e \approx 0.06$

Resolving them needs Δx below about 1 km. Coarser models use this MLE scheme (NEMO, MOM6, CESM).

Fox-Kemper, Ferrari & Hallberg (2008), J. Phys. Oceanogr. 38 · McWilliams (2016), Proc. R. Soc. A 472

PART 5 · HORIZONTAL VISCOSITY

Viscosity: physics and numerical safety valve

LAPLACIAN · BIHARMONIC

$A\,\nabla^2 u$ or $-B\,\nabla^4 u$

biharmonic damps the smallest scales and spares the rest

SMAGORINSKY

$A = (C\,\Delta x)^2\,|D|$

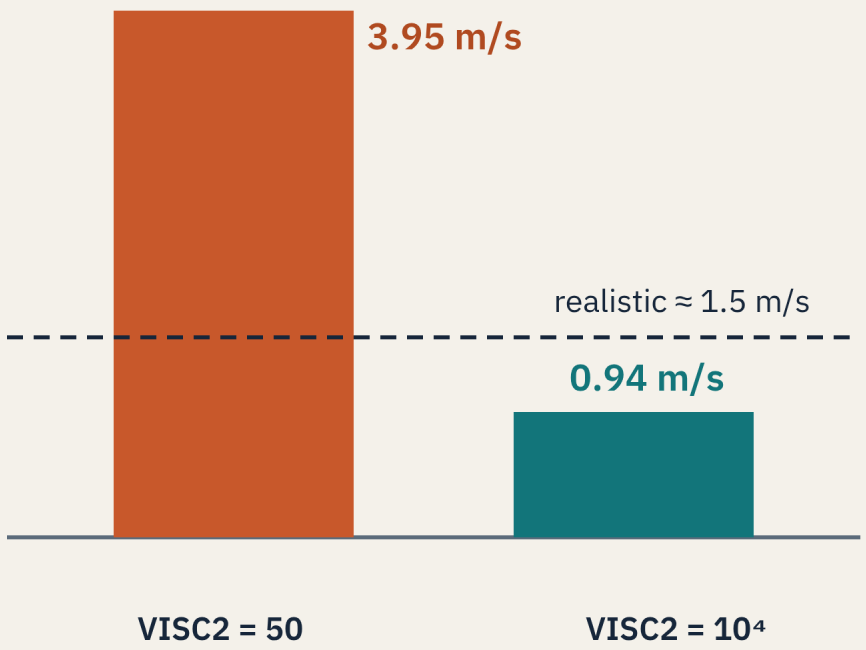
more viscosity only where the flow is strongly deformed

MUNK LAYER MUST SPAN A GRID CELL

$(A/\beta)^{1/3} \geq \Delta x \rightarrow A \gtrsim \beta \Delta x^3 \approx 3 \times 10^4\ m^2/s$

at 1°: $\beta \approx 2.3 \times 10^{-11}\ m^{-1}s^{-1}$, Δx = 111 km

OUR CLASS TEST · MAX SPEED, DAY 1



Grid Reynolds number UΔx/A: 2,200 vs 11

Munk (1950), J. Meteorol. 7 · Smagorinsky (1963), Mon. Wea. Rev. 91 · bars: our 1° runs, Jan 1993 (2nd run also had KPP and the longwave fix)

PART 5 · SUMMARY

What to parameterize depends on Δx

Process	1° · 111 km	1/4° · 28 km	1/12° · 9 km	1/50° · 2 km
Mesoscale eddies	parameterize (GM)	partly resolved	resolved at low latitudes	resolved
Submesoscale fronts	parameterize (MLE)	parameterize (MLE)	parameterize (MLE)	partly resolved
Boundary currents	wide, viscous	too wide	resolved	resolved
Vertical turbulence	KPP / GLS	KPP / GLS	KPP / GLS	KPP / GLS
Horizontal viscosity	large, for stability	moderate	small, biharmonic	small, biharmonic

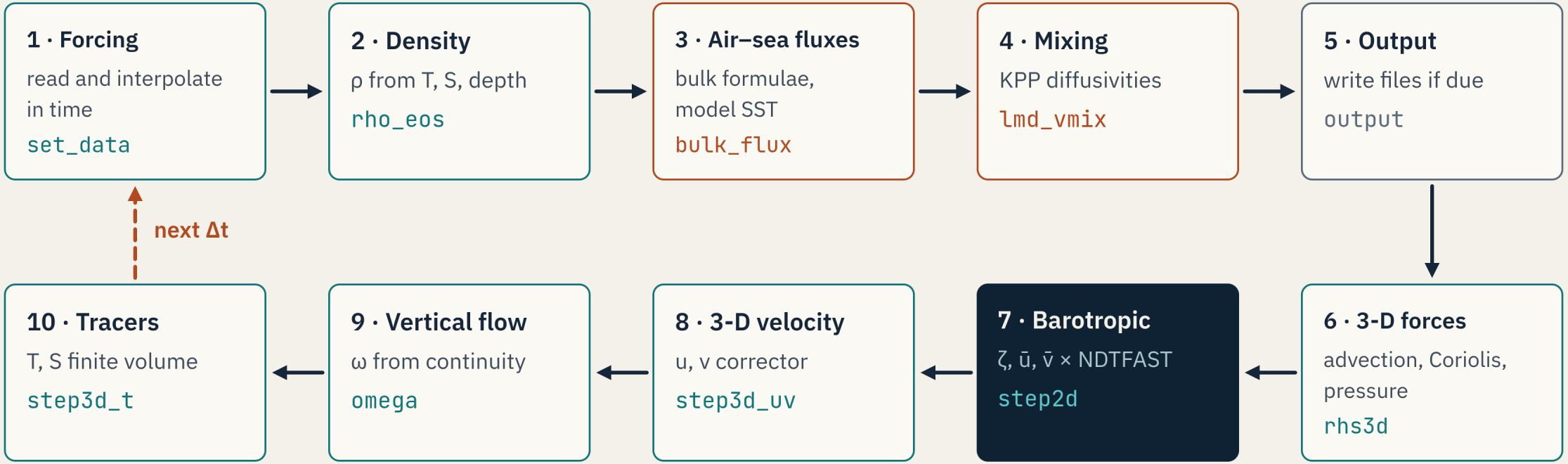
Resolving an eddy needs about two grid points per Rossby radius, so the needed Δx shrinks toward the poles.

Our 1° class box sits in the first column: a teaching tool, not an eddy model.

Hallberg (2013), Ocean Modelling 72 · Fox-Kemper et al. (2019), Front. Mar. Sci. 6: “Challenges and prospects in ocean circulation models”

PART 6 · PUTTING IT TOGETHER

One ROMS time step, in the order the code runs



Parts 2 → 5 all appear here: inputs in 1 and 3, the C-grid and s-levels in 6–9, finite volumes in 10, parameterizations in 4.

Simplified from ROMS/Nonlinear/main3d.F (ROMS revision used in class, Aug 2025)

PART 6 · BEFORE YOU TRUST A RUN

Eight sanity checks, each one learned the hard way

Physics switches
read the activated CPP list at the top of the log

Inputs consistent
same s-levels in IC and .in; source vs model sea floor

Flux signs
net longwave negative, about −50 W/m²

Time reference
DSTART = initial time, or the first average is wrong

CFL and speed
Courant numbers below 1; max speed near 1 m/s, not 4

Energy settles
kinetic energy levels off after spin-up; volume constant

Topography
rx0 below 0.2, rx1 below about 8

Observations
our SST vs GLORYS after a month: bias +0.03 °C, RMS 0.21 °C

Every item here broke at least once while preparing this class.

Model, grids and numerics

Arakawa & Lamb (1977) Methods Comput. Phys. 17, 173–265

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Physics, data and tools

Chelton et al. (1998) J. Phys. Oceanogr. 28, 433–460

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Hallberg (2013) Ocean Modelling 72, 92–103

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Lellouche et al. (2021) Front. Earth Sci. 9, 698876

McWilliams (2016) Proc. R. Soc. A 472, 20160117

Munk (1950) J. Meteorol. 7, 79–93

Redi (1982) J. Phys. Oceanogr. 12, 1154–1158

Smagorinsky (1963) Mon. Wea. Rev. 91, 99–164

Soufflet et al. (2016) Ocean Modelling 98, 36–50

Umlauf & Burchard (2003) J. Mar. Res. 61, 235–265

Web: myroms.org/wiki · roms-tools.readthedocs.io · croco-ocean.org

Class numbers (fluxes, speeds, SST bias) come from our own ROMS runs of the 1° Indian Ocean configuration.

Three things to remember

1

Physics is chosen when you compile. Read the log to see what you got.

2

Most failures come from inputs, not from the model code.

3

Every scheme and parameterization is a trade-off: check against observations.

Try it: ROMS_class_2026.ipynb · ROMS_input_preparation.ipynb

Questions?